Squares, Cubes & Roots

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Squares, cubes & roots

What are square numbers?

- A **square** number is the result of multiplying a number by itself
  - The first square number is $1 \times 1 = 1$, the second is $2 \times 2 = 4$ and so on
- The first 15 square numbers are: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225
  - Aim to remember at least the first fifteen square numbers
- In algebra, square numbers can be written using a power of 2
  - $a \times a = a^2$

What are cube numbers?

- A **cube** number is the result of multiplying a number by itself, twice
  - The first cube number is $1 \times 1 \times 1 = 1$, the second is $2 \times 2 \times 2 = 8$ and so on
- The first 5 cube numbers are 1, 8, 27, 64 and 125
  - Aim to remember at least the first five cube numbers
  - You should also remember $10^3 = 1000$
- In algebra, cube numbers can be written using a power of 3
  - $a \times a \times a = a^3$

What are square roots?

- The **square root** of a value, is the number that when multiplied by itself equals that value
  - For example, 4 is the square root of 16
  - It is the inverse of squaring
  - Square roots are indicated by the symbol $\sqrt{\ }$
    - e.g. The **positive** square root of 49 would be written as $\sqrt{49}$
  - Square roots can be positive and negative
    - e.g. The square roots of 25 are 5 and -5
  - If a negative square root is required then a - sign would be used

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

$\pm\sqrt{25}$

- The square root of a non-square **integer** is also called a **surd**
  - e.g. $\sqrt{3}$ is a surd, as 3 is not a square number
  - surds are **irrational** numbers
    - where possible modern calculators will display **irrational** numbers as surds
  - $\sqrt{64}$ is **rational,** as it is equal to 8
    - 64 is a square number
  - However, $\sqrt{2}$ is **irrational**
    - 2 is not a square number
- You should aim to remember the square roots of the first 15 square numbers:
  - $\sqrt{1}, \sqrt{4}, \sqrt{9}, \sqrt{16}, \sqrt{25}, \sqrt{36}, \sqrt{49}, \sqrt{64}, \sqrt{81}, \sqrt{100},$

What are cube roots?

- The **cube root** of a value, is the number that when multiplied by itself twice equals that value
  - For example, 3 is the cube root of 27
  - It is the inverse of cubing
  - Cube roots are indicated by the symbol $\sqrt[3]{}$
    - e.g. The cube root of 64 would be written as $\sqrt[3]{64}$
  - You should remember the values of the following cube roots:
    - $\sqrt[3]{1}, \sqrt[3]{8}, \sqrt[3]{27}, \sqrt[3]{64}, \sqrt[3]{125}, \sqrt[3]{1000}$

Worked Example

Write down a number which is both a cube number and a square number, and hence express this number in two different ways using powers of 2 and 3.

Answer:

Listing the first 12 square numbers

1, 4, 9, 16, 25, 36, 49, **64**, 81, 100, 121, 144

Listing the first 5 cube numbers

1, 8, 27, **64**, 125

64 appears in both lists, it is the 8th square number and 4th cube number

64 is both a square and cube number
$64 = 8^2$ and $64 = 4^3$

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